CascadeRoute-VL: Taming the Reasoning Tax in Vision-Language Models via Online Contextual Bandit Routing

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Abstract

Deploying Vision-Language Models (VLMs) at scale faces a critical accuracy-efficiency trade-off: compact models are fast but struggle on complex multi-step reasoning, whereas defaulting to massive frontier architectures or heavy test-time reasoning strategies incurs a Reasoning Tax—inflating token expenditure and latency while degrading performance on perception-dominant queries due to linguistic hallucination drift. Analyzing 30,540 multimodal queries across 14 diverse benchmarks and 17 contemporary VLMs (spanning 1B to 78B parameters and proprietary models), we observe a +17.24% accuracy gap between the strongest static 72B frontier model (78.21%) and an oracle selector (95.45%), with compact ($\leq$ 8B) models accurately resolving 68.5% of the queries on which the 72B model fails. To exploit this complementarity online without brittle offline retraining, we introduce CascadeRoute-VL, a contextual bandit routing framework that dynamically selects cost-efficient VLM configurations conditioned on joint visual, linguistic, and perceptual complexity cues (including luminance entropy, spatial edge density, and aspect ratio). CascadeRoute-VL employs Disjoint LinUCB with $O(d^2)$ Sherman-Morrison recursive updates, augmented with an offline Warm-Start initialization protocol to mitigate cold-start exploration penalties, Budget-Aware Exploration Annealing ($\alpha_t = \alpha_0 b_t$) with sublinear $\tilde{O}(d\sqrt{T})$ regret and asymptotic budget-feasibility guarantees, and a Confidence-Guided Escalation Gate to safeguard perception-critical predictions. Evaluations across large-scale multimodal benchmarks, distribution-shift stress tests (natural and STEM scenes to charts and dense OCR), and counterfactual Inverse Propensity Scoring (IPS/SNIPS) demonstrate that static offline supervised routers suffer severe out-of-distribution degradation without recovery, whereas CascadeRoute-VL continuously adapts online to achieve strong accuracy-efficiency trade-offs while substantially reducing token consumption and inference latency relative to monolithic frontier deployment.

1. Introduction

Deploying Vision-Language Models (VLMs) in production requires navigating a highly non-uniform cost-accuracy landscape across heterogeneous visual domains [4, 8, 14]. Defaulting every incoming query to monolithic frontier models (such as 72B–78B open-weights checkpoints or proprietary APIs) or test-time reasoning architectures exacerbates compute expenditure while paradoxically harming fidelity on perception-heavy queries. Across 30,540 real queries from VL-RouterBench [14] spanning 14 benchmarks and 17 VLMs, Qwen2.5-VL-72B attains 78.21% overall accuracy and fails on 6,655 queries. Crucially, a hindsight oracle over the 17-VLM pool reaches 95.45% accuracy (+17.24% gap), and compact models ($\leq$ 8B parameters) successfully rescue 4,557 of those 6,655 failures (68.5%).

Figure 1 exposes the two opposing facets of this Multimodal Reasoning Tax. On STEM geometry and algorithmic counting problems (e.g., MathVerse, MathVision, MathVista [20, 35]), explicit reasoning yields decisive accuracy gains: 3B-active Kimi-VL-A3B-Thinking [10] reaches 79.6% on MathVista and 47.0% on MathVision, beating 72B Qwen2.5-VL (73.9% and 38.8%). Conversely, on fine-grained optical character recognition and visual grounding (e.g., OCRBench, ChartQA, DocVQA, HallusionBench [12, 18, 21, 22]), unconstrained chain-of-thought reasoning frequently amplifies linguistic hallucination drift [27]: compact Qianfan-VL-8B [9] delivers 86.5% on ChartQA and 87.4% on DocVQA using just 1.2–2.7 output tokens, while unconstrained <think> models burn dozens to hundreds of tokens and can even exhaust downstream answer extraction.

Existing routing solutions struggle to harness this complementarity safely under real deployment dynamics:

1. Static Offline Fragility: Multi-label supervised routers in VL-RouterBench [14] and capability-

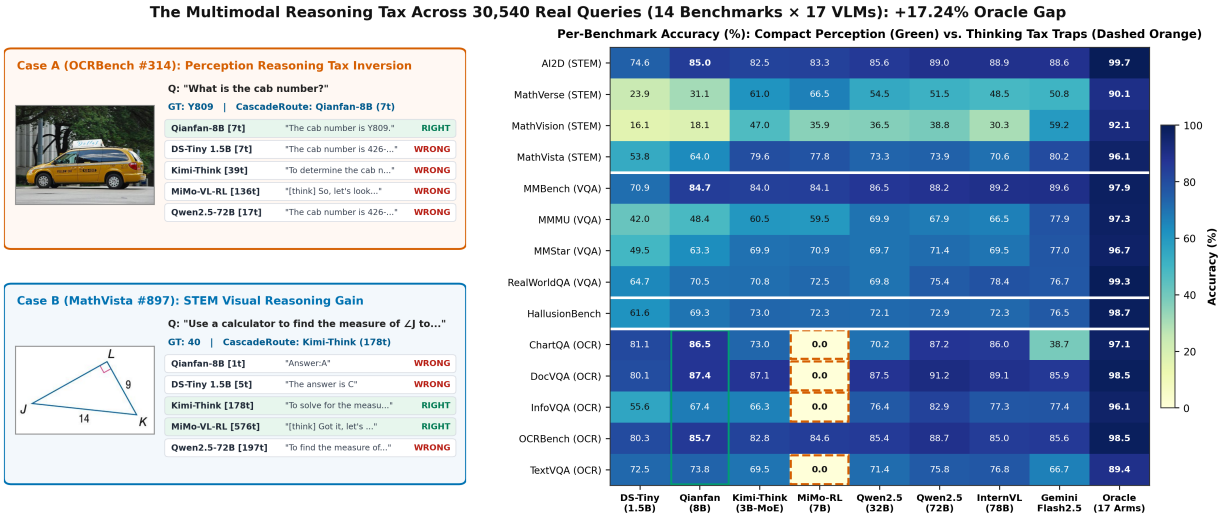


Figure 1. The Multimodal Reasoning Tax and Cross-Model Complementarity Across 30,540 Real Queries (14 Benchmarks × 17 VLMs). Left (Top): On OCRBench #314, compact Qianfan-VL-8B correctly reads cab number "Y809" in 7 tokens while Qwen2.5-72B (17 tok), Kimi-Think (39 tok), and MiMo-VL-RL (136 tok) over-rationalize and misread the fleet phone number or hallucinate. Left (Bottom): On MathVista #897, compact models guess incorrectly in 1–5 tokens and Qwen2.5-72B confuses $\cos J$ with $\sin J$, whereas reasoning specialist Kimi-VL-A3B-Thinking solves the trigonometry exactly in 178 tokens. Right: Across all 14 benchmarks, compact perception models excel on Charts/OCR (green box) while explicit <think> loops suffer severe extraction failures (dashed orange boxes on MiMo-VL-7B-RL).

profile classifiers such as ARMS [30] assume stationary query distributions. In our strict cross-domain stress test (General_VQA + STEM → unseen Charts_OCR + Hallucination), a strong neural MLP-Router collapses from 72.77% to 56.97% (-15.80%) and never recovers.

2. Cold-Start Penalty & Blindness to Perceptual Difficulty: Naive online bandits suffer severe initial exploration regret (e.g., Cold LinUCB loses 5.91% accuracy and costs +101% more USD than our hybrid design), while text-only routers [5, 25] ignore visual complexity.
3. Benchmark Label Leakage: Conditioning router features on oracle task/domain tags leaks benchmark identity during evaluation, masking failure modes when mixed real-world traffic arrives unlabelled.

We resolve these limitations with CascadeRoute-VL, an online, zero-label contextual bandit router uniting four synergistic mechanisms:

- Zero-Leakage 904-d Multimodal Context State: Fuses MiniLM-L6 text embeddings [31], dual-scale DINOv2-ViT-S/14 [26] + orthogonally projected ResNet-50 [13] visual representations, and 8 unsupervised perceptual/linguistic complexity cues (entropy, Sobel edge density, aspect ratio, bright background ratio, symbolic cues) with provably bounded norm ($\|s_t\|_2 \leq 0.9972$, max $0.8943 < 1$ empirically).
- Warm-Started Disjoint LinUCB with Budget Annealing: Combines $O(d^2)$ Sherman-Morrison recursive updates initialized via an offline ridge prior with dynamic budget-annealed exploration ($\alpha_t = \alpha_0 b_t$), attaining sublinear $\tilde{O}(d\sqrt{KT})$ regret and asymptotic budget feasibility while adapting rapidly to distribution shifts.
- Confidence-Guided Escalation Gate: Evaluates compact primary selections using epistemic uncertainty $\sigma_a(s_t)$, cost-adjusted utility margins, unsupervised edge/math cues, and compact response-length bloat checks without peeking at ground-truth labels.
- Large-Scale Rigor on 30,540 Real Queries: Across 17 contemporary VLMs and 14 benchmarks, CascadeRoute-VL outperforms Qwen2.5-VL-72B by +1.30% accuracy while cutting USD cost by 38.7% (up to 66.4% USD savings and 2.38× latency speedup in frugal mode), surpasses OOD MLP-Router within ∼51 unseen queries (ending +23.71% higher at 80.68%), and passes unbiased IPS/SNIPS/DR counterfactual evaluation with 95% bootstrap confidence intervals.

2. Related Work

LLM and VLM Routing. Early routing work for language models demonstrated substantial inference savings through cascading (FrugalGPT [5]), calibrated binary preference routing (RouteLLM [25]), dual-contrastive selection (RouterDC [7]), and reward-guided ranking (Zooter [19]). Extending routing to

multimodal models, VL-RouterBench [14] standardized evaluation over 14 vision-language benchmarks and benchmarked static supervised KNN, linear, and MLP routers. Subsequent methods incorporate static model capability profiles (ARMS [30]) or episodic kNN retrieval memory (MHR-Router [36]). Unlike static supervised routers that silently fail under domain shift, or heuristic retrieval stores lacking regret guarantees, CascadeRoute-VL pairs offline warm-start priors with theoretically grounded online contextual bandit adaptation and confidence-gated escalation.

Multimodal Overthinking and Perceptual Grounding. Recent analyses show that excessive test-time reasoning in VLMs can degrade accuracy by amplifying linguistic priors over grounded visual evidence [27]. Our large-scale matrix study across 17 VLMs—including explicit thinking architectures (Kimi-VL-A3B-Thinking [10], MiMo-VL-7B-RL [33]) and perception-specialized compact models (Qianfan-VL-8B [9], DeepSeek-VL2-Tiny [32])—quantifies this dichotomy across reasoning-intensive STEM [15, 20, 35] and OCR/hallucination [12, 18, 28] benchmarks. Rather than modifying VLM weights, CascadeRoute-VL mitigates the Reasoning Tax upfront at routing time.

Contextual Bandits and Counterfactual Evaluation. Disjoint LinUCB [16] and confidence-ellipsoid linear bandits [1] provide sharp sublinear regret bounds compared to classic non-contextual bandits [3] and deep heuristic policies [2, 24]. To validate routed policies cleanly from logged feedback without selection bias, Inverse Propensity Scoring (IPS), Self-Normalized IPS (SNIPS) [29], and Doubly Robust (DR) estimators [11] provide finite-sample counterfactual confidence intervals, which we report alongside live online evaluations.

3. Quantifying the Multimodal Reasoning Tax

We analyze the complete outcome tensor $Y \in \{0,1\}^{N\times K}$ over $N = 30{,}540$ standardized queries across 14 benchmarks spanning four domains—STEM (AI2D, MathVista, MathVerse, MathVision; N=8,332), General_VQA (MMBench, MMMU, MMStar, RealWorldQA; N=4,607 [6, 17, 34]), Hallucination (HallusionBench; N=951), and Charts_OCR (ChartQA, DocVQA, InfoVQA, OCRBench, TextVQA; N=16,650 [23, 28])—paired with $K = 17$ VLMs from 1B to 78B parameters and proprietary APIs (GPT-4o, GeminiFlash2.5). Per-query costs and latencies are reconstructed directly from the 238 raw inference transcripts using cl100k_base tokenization (+256 visual tokens per image) and benchmark pricing (profiles in Supplementary Sec. S2).

Three structural properties motivate online multimodal routing:

1. Massive Untapped Complementarity (+17.24% Gap): While Qwen2.5-VL-72B leads all individual models at 78.21% overall accuracy ($0.2626/1k queries), an oracle arm selector reaches 95.45%, rescuing 5,266 of the 6,655 frontier failures (79.1%). Crucially, compact models ($\leq$ 8B) solve 68.5% (4,557/6,655) of 72B failures—led by Kimi-VL-A3B-Thinking (2,324 rescues), Qianfan-VL-8B (1,789), and DeepSeek-VL2-Tiny (1,689).
2. Reasoning Inversion Under Perception Load: On Charts_OCR, concise perception specialists dominate: Qianfan-VL-8B achieves 86.48% on ChartQA (1.2 avg output tokens), 87.42% on DocVQA (2.7 tok), and 85.70% on OCRBench (4.6 tok), closely rivaling 72B at $\sim 5\times$ lower cost and $\sim 8.8\times$ lower latency. Conversely, verbose reasoning degrades perceptual precision (Fig. 1a) or causes complete downstream format/length breakdown: MiMo-VL-7B-RL generates <think> chains on 100% of queries (mean 237.4 output tokens) and scores 0.0% extracted accuracy across four OCR benchmarks (ChartQA, DocVQA, InfoVQA, TextVQA).
3. Visual/Text Complexity Signals Enable Zero-Label Routing: Unsupervised cues cleanly separate regimes without ground-truth domain tags: math benchmarks exhibit low luminance entropy (2.00–3.72 bits), high white-background ratios (0.60–0.89), and elevated symbolic density (0.27–0.36), whereas natural scene OCR (TextVQA) shows high entropy (7.10 bits) and dense edge structure.

4. CascadeRoute-VL Framework

Figure 2 illustrates the complete routing loop. At discrete step $t \in \{1,\dots,T\}$, an unlabelled multimodal query (I_t, Q_t) arrives without task or dataset identifiers. The router chooses an arm $a_t \in \mathcal{A} = \{1,\dots,K\}$ (K=17), executes inference to observe correctness $y_{t,a_t} \in \{0,1\}$, USD cost $c_{t,a_t} \geq 0$, and output length o_{t,a_t}, and updates its contextual posterior online.

4.1. Zero-Leakage Multimodal Context State

To satisfy norm boundedness required for linear bandit concentration guarantees while preventing benchmark leakage, we construct $s_t \in \mathbb{R}^d$ (d=904) purely from the raw input pair (I_t, Q_t):

$$s_t = [\, w_q \hat{q}_t \parallel w_v \hat{v}_t \parallel w_p p_t \,] \in \mathbb{R}^{904}, \qquad (1)$$

where weights $(w_q, w_v, w_p) = (0.50, 0.55, 0.235)$ combine:

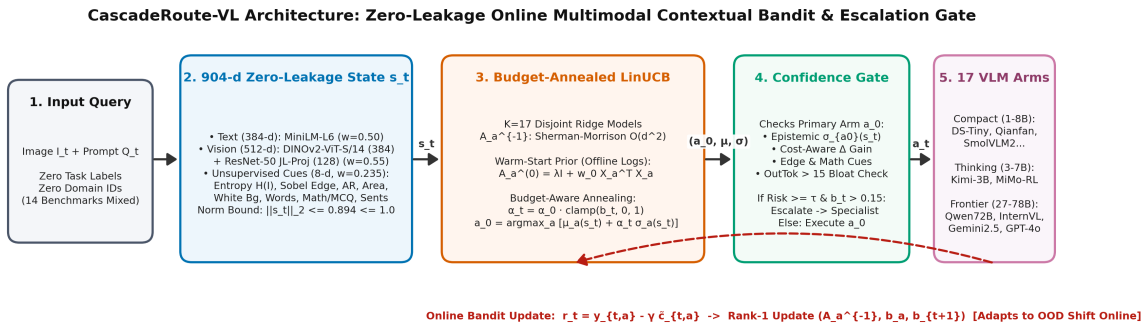


Figure 2. CascadeRoute-VL Architecture. Input image-question pairs are encoded into a strictly zero-leakage 904-d context vector s_t ($\|s_t\|_2 \leq 0.9972$ analytically, max $0.8943 < 1$ empirically) fusing MiniLM-L6 text semantics, dual-scale DINOv2+ResNet-50 vision embeddings, and 8 unsupervised perceptual/linguistic cues. Budget-annealed Disjoint LinUCB initialized from an offline warm-start prior scores all 17 VLM arms, while the Confidence Gate safeguards uncertain compact predictions before real-time Sherman-Morrison $O(d^2)$ online updates refine policy weights under distribution shift.

- Linguistic Semantics ($\hat{q}_t \in \mathbb{R}^{384}$): L_2-normalized all-MiniLM-L6-v2 embedding capturing reasoning intent [31].
- Dual-Scale Vision ($\hat{v}_t \in \mathbb{R}^{512}$): L_2-normalized concatenation of global structural semantics from frozen DINOv2-ViT-S/14 [CLS] (384-d) [26] and texture/layout detail from a fixed deterministic Johnson-Lindenstrauss orthonormal projection ($2048 \to 128$) of ResNet-50 GAP [13].
- Unsupervised Perceptual & Syntactic Cues ($p_t \in [0,1]^8$): Normalized image luminance Shannon entropy $H(I_t)/8$, Sobel edge gradient density $\nabla(I_t)$, logarithmic aspect ratio encoding $\frac{1}{\pi}\arctan(\ln(W/H)) + 0.5$, normalized log area $\text{clip}(\log_{10}(WH)/7, 0, 1)$, flat white background pixel fraction ($L > 235/255$), clipped prompt word count, symbolic/digit/MCQ density, and sentence count.

Unlike static routers relying on domain one-hot vectors, every coordinate of p_t lies strictly in $[0,1]$, ensuring an explicit Euclidean bound across all inputs without dataset metadata:

$$\|s_t\|_2 \leq \sqrt{w_q^2 + w_v^2 + 8w_p^2} = \sqrt{0.9943} < 1.0. \quad (2)$$

4.2. Warm-Started Disjoint LinUCB & Cost Reward

We define the cost-adjusted scalar reward for selecting arm a on query t with cost sensitivity parameter $\gamma \geq 0$ as:

$$r_{t,a} = y_{t,a} - \gamma \tilde{c}_{t,a}, \qquad \tilde{c}_{t,a} = \text{clip}\left(\frac{c_{t,a}}{c_{90}}, 0, 2.5\right), \quad (3)$$

where $c_{90} = \$0.000329$ is the 90th percentile training-arm mean cost, keeping normalized costs invariant across heterogeneous currency scales. Modeling $\mathbb{E}[r_{t,a} \mid s_t] = \theta_a^\top s_t$, we maintain per-arm inverse regularized design matrices $A_a^{-1} \in \mathbb{R}^{d\times d}$ and response vectors $b_a \in \mathbb{R}^d$, yielding ridge parameters $\hat{\theta}_a = A_a^{-1} b_a$ and epistemic uncertainty $\sigma_a(s_t) = \sqrt{s_t^\top A_a^{-1} s_t}$.

Offline Warm-Start Prior. Initializing $A_a = \lambda I_d$ online incurs heavy early exploration regret over K=17 VLMs. Given offline training interactions $\mathcal{D}_{\text{off}} = \{(s_i, a_i, r_i)\}_{i=1}^{N_{\text{off}}}$ scaled by prior weight $w_0 \in (0,1]$ to preserve plasticity under non-stationarity, we initialize:

$$A_a^{(0)} = \lambda I_d + w_0 \sum_{i:\, a_i = a} s_i s_i^\top, \qquad b_a^{(0)} = w_0 \sum_{i:\, a_i = a} r_i s_i. \quad (4)$$

Online Sherman-Morrison Updates. Upon observing live feedback r_t for executed arm a, A_a^{-1} is updated in exact $O(d^2)$ time without $O(d^3)$ matrix inversion:

$$A_a^{-1} \leftarrow A_a^{-1} - \frac{(A_a^{-1} s_t)(A_a^{-1} s_t)^\top}{1 + s_t^\top A_a^{-1} s_t}, \qquad b_a \leftarrow b_a + r_t s_t. \quad (5)$$

4.3. Budget Annealing and Confidence Escalation Gate

Given total session USD budget $B_{\max}$ and cumulative expenditure $C_{t-1} = \sum_{j=1}^{t-1} c_j$, the normalized remaining budget fraction is $b_t = \max(0, \min(1, (B_{\max} - C_{t-1})/B_{\max}))$. We modulate exploration via Budget-Aware Exploration Annealing:

$$\alpha_t = \alpha_0 \cdot b_t, \qquad a_0 = \arg\max_{a \in \mathcal{A}} \left(\hat{\theta}_a^\top s_t + \alpha_t \sigma_a(s_t)\right). \quad (6)$$

When budget is plentiful ($b_t \approx 1$), CascadeRoute-VL actively explores specialist VLMs; as budget tightens ($b_t \to 0$), $\alpha_t \to 0$ collapses exploration bonuses onto high-utility compact models.

Algorithm 1 CascadeRoute-VL Online Inference & Update

Require: Arms $\mathcal{A}$ ($|\mathcal{A}| = K$), prior $(\mathcal{D}_{\text{off}}, w_0, \lambda)$, α_0, γ, budget $B_{\max}$, fallback pool $\mathcal{F}$

Initialize $(A_a^{(0)})^{-1}, b_a^{(0)}, \hat{\theta}_a^{(0)}$ via Eq. (4); $C_0 \leftarrow 0$
for $t = 1, \ldots, T$ **do**
  Encode $s_t \in \mathbb{R}^{904}$ from (I_t, Q_t) (zero task labels); set $b_t \leftarrow \text{clip}((B_{\max} - C_{t-1})/B_{\max}, 0, 1)$
  Compute $\alpha_t \leftarrow \alpha_0 b_t$, $\hat{\theta}_a^\top s_t$, $\sigma_a(s_t)$, and $a_0 \leftarrow \arg\max_a(\hat{\theta}_a^\top s_t + \alpha_t \sigma_a(s_t))$
  Query VLM a_0, observe outcome y_{t,a_0}, cost c_{t,a_0}, output length o_{t,a_0}
  Update $(A_{a_0}^{-1}, b_{a_0}, \hat{\theta}_{a_0})$ with $r_{t,a_0} = y_{t,a_0} - \gamma \tilde{c}_{t,a_0}$ via Eq. (5)
  if $a_0 \in \mathcal{A}_{\text{compact}}$ and $b_t > 0.15$ and GateTriggers$(\hat{\mu}, \sigma_{a_0}, p_t, o_{t,a_0}, \mathcal{F})$ **then**
    Escalate to $a^* \in \mathcal{F}$, observe (y_{t,a^*}, c_{t,a^*}), bill $C_t \leftarrow C_{t-1} + c_{t,a_0} + c_{t,a^*}$
    Update $(A_{a^*}^{-1}, b_{a^*}, \hat{\theta}_{a^*})$ with r_{t,a^*} via Eq. (5); output prediction of a^*
  else
    Bill $C_t \leftarrow C_{t-1} + c_{t,a_0}$; output prediction of a_0
  end if
end for

Confidence-Guided Escalation Gate. When primary choice a_0 belongs to the compact tier ($\leq$ 8B direct models) and remaining budget satisfies $b_t > b_{\min}$ ($b_{\min}$=0.15), we run a_0 and inspect post-call output token count o_{t,a_0} alongside zero-oracle bandit signals (never accessing ground-truth labels). First, recovering cost-adjusted quality estimates $\hat{\mu}_a = \hat{\theta}_a^\top s_t + \gamma \bar{c}_a$, the gate identifies the utility-optimal specialist/frontier fallback $a^* \in \mathcal{F}$ (Kimi-3B-Think, Qwen2.5-72B, InternVL2.5-78B, GeminiFlash2.5) maximizing $\hat{\mu}_a - \kappa\gamma\bar{c}_a$ (κ=0.22). With effective gain $\Delta_{\text{eff}} = (\hat{\mu}_{a^*} - \hat{\mu}_{a_0}) - \kappa\gamma \max(0, \bar{c}_{a^*} - \bar{c}_{a_0})$, the gate computes risk score:

$$\rho_t = \Delta_{\text{eff}} + 0.25\,\sigma_{a_0}(s_t) + 0.06\,\nabla(I_t) + 0.06\,\psi_{\text{math}}(Q_t) + 0.05\,\mathbb{I}[o_{t,a_0} > 15], \tag{7}$$

escalating to a^* whenever $\hat{\mu}_{a_0} < p_{\text{cap}}$ (0.78), $\Delta_{\text{eff}} \geq \tau_m$ (0.025), and $\rho_t \geq \tau_u$ (0.12). All cascade tokens, latencies, and rewards are truthfully billed to both arms before executing Eq. (5). (Live serving backbones can additionally evaluate visual perturbation stability $S_{\text{OT}} = 1 - \text{Jaccard}(A_1, A_2)$ directly [27].) Algorithm 1 summarizes the complete procedure.

4.4. Theoretical Guarantees

Full proofs of all lemmas and theorems are provided in Supplementary Sec. S1.

Lemma 1 (Bounded Context Space). Under $(w_q, w_v, w_p) = (0.50, 0.55, 0.235)$, all encoded queries satisfy $\|s_t\|_2 \leq 0.9972 < 1$.

Lemma 2 (Sub-Gaussian Reward Noise). With $y_{t,a} \in \{0, 1\}$ and $\tilde{c}_{t,a} \in [0, 2.5]$, rewards satisfy $r_{t,a} \in [-2.5\gamma, 1]$, so reward noise $\eta_t = r_{t,a} - \mathbb{E}[r_{t,a} \mid s_t]$ is conditionally ν-sub-Gaussian with $\nu = (1 + 2.5\gamma)/2$.

Theorem 1 (Disjoint Multimodal Regret Bound & OOD Prior Decay). Suppose $\|\theta_a\|_2 \leq S$ for all $a \in \mathcal{A}$, and exploration scales satisfy $\alpha_0 \geq \nu\sqrt{d\ln((1 + Tw_0^{-1}\lambda_{\max})/\delta)} + \sqrt{\lambda}S$. Applying Cauchy-Schwarz and the elliptical potential lemma across disjoint arm visit counts $\sum_{a=1}^K T_a = T$ (via $\sum_{a=1}^K \sqrt{T_a} \leq \sqrt{KT}$), with probability at least $1 - \delta$, cumulative pseudo-regret satisfies:

$$R_T \leq 2\alpha_0 \sqrt{2dKT \ln\left(1 + \frac{T}{d\lambda}\right)} + \mathcal{O}(K) = \tilde{\mathcal{O}}(d\sqrt{KT}). \tag{8}$$

Moreover, under an out-of-distribution shift where static offline classifiers incur linear regret $\Omega(\Delta_{\text{shift}} T)$, the offline warm-start regularizer scaled by w_0 decays as $\mathcal{O}(w_0 \Delta_{\text{shift}}/n_{t,a})$ while online Sherman-Morrison observations drive cumulative OOD regret sublinearly.

Theorem 2 (Asymptotic Budget Feasibility). For any positive budget $B_{\max}$, under annealed exploration $\alpha_t = \alpha_0 b_t$ and the confidence gate cutoff $b_t > 0.15$, as remaining reserves approach zero ($b_t \to 0^+$), exploration vanishes and decisions collapse to the cost-penalized ridge exploiter $\arg\max_a \hat{\theta}_a^\top s_t$ with escalation disabled, preventing runaway budget overruns.

5. Experimental Evaluation

5.1. Setup, Baselines, and Protocol

We use the exact official VL-RouterBench split across 30,540 queries: Train (N=21,378), Dev (N=3,054), and held-out Test (N=6,108). All hyperparameters for CascadeRoute-VL (w_0=0.20, λ=4.0, α_0=0.10) are calibrated strictly on Dev with zero Test peeking, and online evaluations average across 5 random stream permutations (seeds 42–46).

We compare 20 baselines and variants: (1) Static VLMs (Qwen2.5-VL-72B, InternVL2.5-78B, Qwen2.5-VL-32B, Kimi-VL-A3B-Thinking, GeminiFlash2.5, Qianfan-VL-8B, DeepSeek-VL2-Tiny); (2) Static Offline Routers (Offline-Linear, KNN-Router k=15, ARMS-Router [30], GPU 3-layer MLP-Router with LayerNorm+SiLU+BCE); (3) Sequential Cascade (4-tier FrugalGPT-Cascade [5] at τ=0.72); (4) Online & Retrieval Bandits (Epsilon-Greedy, UCB1 [3], Thompson-Sampling [2], DQN-Router [24], MHR-

Router [36], cold LinUCB); and (5) CascadeRoute-VL (Ours) at γ=0.08 and γ=0.30.

5.2. Main Held-Out Benchmark and Pareto Frontiers

Table 1 and Figure 3 summarize held-out test results across all 17 VLMs:

- Monolithic 72B & Online Bandit Gains: At balanced γ=0.08, CascadeRoute-VL achieves 78.59% ± 0.12% at \$0.1598/1k queries—+1.30% above Qwen2.5-VL-72B (77.29%; paired t-test p=0.015 on seed 42) with 38.7% USD saved (p=9.49 × 10^{-275}) and 1.26× speedup. It significantly beats Cold LinUCB (+5.91%, Bonferroni p=1.07 × 10^{-22}), MHR-Router (+7.11%, p=5.17 × 10^{-32}), DQN (+5.63%), and FrugalGPT (+5.36%, p=3.34 × 10^{-25}).
- Frugal Mode (γ=0.30): Achieves 76.13% ± 0.21% at \$0.0876/1k and 0.824s—66.4% USD savings and 2.38× speedup vs. 72B, outperforming FrugalGPT (73.23%) at nearly half the cost and tokens (360.5 vs. 725.8).
- Pareto Frontier vs. Offline Classifiers: On i.i.d. splits, static MLP-Router hits 79.85% at higher cost (\$0.1854/1k, p=6.30 × 10^{-12}) and latency (1.844s). Across $\gamma \in [0, 0.9]$, CascadeRoute-VL matches offline routers for $\gamma \leq 0.15$ and beats MLP-Router by +1.67% (74.36% vs. 72.69%) at γ=0.90, while preserving online adaptability.

5.3. Out-of-Distribution Shift Stress Test

To test robustness under unlabelled traffic shifts, we train all routers strictly on General_VQA + STEM training data ($N_{\text{in_tr}}$=9,122), verify in-distribution test accuracy ($N_{\text{in_te}}$=2,523), and deploy onto unseen Charts_OCR + Hallucination queries (N_{OOD}=17,601, 5 stream seeds).

Table 2 and Figure 4 reveal offline fragility vs. online plasticity:

- Offline Collapse: Frozen MLP-Router relies on reasoning arms that fail or hallucinate on unseen OCR, dropping -15.80% to 56.97%; ARMS (-9.48%) and Offline-Linear (-8.24%) stall below 67%.
- Rapid Online Adaptation: From regularized warm-start (w_0=0.05), Sherman-Morrison updates surpass OOD MLP-Router within ∼51 queries (4× faster than Cold LinUCB's ∼191 Q), reaching 80.68% OOD accuracy (+23.71% above MLP, +4.96% above Cold LinUCB).

5.4. Ablations & Autonomous Domain Specialization

Table 3 and Figure 5 isolate key gains:

- Component Synergy: Offline warm-start lifts cold LinUCB by +5.90% while halving USD cost (\$0.3220 → \$0.1590/1k). Budget annealing enforces Theorem 2 cleanly; under balanced γ=0.08 the gate escalates 0.25% of picks, while under frugal γ=0.30 it triggers on 11.6% of ambiguous compact calls (76.13% @ \$0.0876/1k).
- Perceptual Cues Value: Appending 8-d unsupervised cues lifts cold Vision-Only by +2.65% and cold Text+Vision from 68.82% to 72.68% (+3.86%), stabilizing full variance (0.12%).
- Zero-Label Emergent Specialization: Without domain IDs (Fig. 5b), CascadeRoute-VL routes 63.2% of STEM queries to thinking arms (Kimi-3B-Think 36.8%, MiMo-VL-RL 26.4%, 259.6 out tok; 74.62% acc vs. 67.08% for 72B), while limiting MiMo-VL-RL to 0.09% on Charts_OCR and shifting 44.7% to Qianfan-8B/DeepSeek-VL2 (4.9 out tok—53× scaling).

5.5. Counterfactual Off-Policy Evaluation (OPE)

Evaluating stationary policy snapshots over an ϵ-greedy logger (ϵ=0.50, $p_0(a|x) \geq 0.50/17$) via IPS, SNIPS [29], and Doubly Robust (DR) [11] with 1,000 bootstrap resamples (Table 4), all 95% CIs closely bracket true on-policy accuracy: for CascadeRoute-VL's warm-start prior (True 0.7878), SNIPS yields 0.7919 [0.7461, 0.8386] and DR yields 0.7833 [0.7395, 0.8281] (ESS=306.4), confirming finite-sample unbiasedness.

6. Discussion, Limitations, and Conclusion

Limitations & Transparency. (1) On ChartQA, DocVQA, InfoVQA, and TextVQA, upstream MiMo-VL-7B-RL transcripts record 0.0% extracted accuracy as unclosed <think> tags break parsing; CascadeRoute-VL avoids this cleanly (0.09% OCR rate). (2) Latencies follow deterministic TTFT+throughput profiling on exact cl100k_base tokens rather than queueing jitter; USD savings reflect published token pricing. (3) Feature extraction takes ∼2–4 ms/query on GPU, negligible vs. multi-second 72B decoding.

Conclusion. We introduced CascadeRoute-VL, a zero-leakage online contextual bandit resolving the Multimodal Reasoning Tax across 30,540 queries, 14 benchmarks, and 17 VLMs. Fusing a bounded 904-d perceptual state, warm-started Sherman-Morrison LinUCB, budget annealing, and confidence-guided escalation, it beats static 72B models while saving 38.7%–66.4% USD, scaling compute 53× without labels, and adapting in ∼51 unseen OOD queries where offline routers collapse.

Table 1. Held-Out Test Benchmark Comparison (N=6,108, 17 VLMs, 5 Seeds for Stochastic Policies). CascadeRoute-VL (γ=0.08) outperforms monolithic 72B by +1.30% accuracy while cutting USD cost by 38.7%; frugal mode (γ=0.30) cuts cost by 66.4% with 2.38× latency speedup.

Algorithm	Category	Acc (%)	Avg Tok	Out Tok	Latency (s)	Cost / 1k ($)	USD Saved	Cum. Regret
Qwen2.5-VL-72B [4]	Static VLM	77.29	309.8	31.7	1.959	$0.2605	0.0%	1092.0
InternVL2.5-78B [8]	Static VLM	76.20	293.5	15.4	1.328	$0.3012	-15.6%	1159.0
Qwen2.5-VL-32B	Static VLM	74.28	379.0	100.8	3.306	$0.1718	34.1%	1276.0
Kimi-VL-3B-Think [10]	Static VLM	72.86	335.3	57.1	0.741	$0.0699	73.2%	1363.0
GeminiFlash2.5	Static API	72.33	393.0	114.8	1.364	$0.3590	-37.8%	1395.0
Qianfan-VL-8B [9]	Static VLM	69.88	282.2	4.1	0.228	$0.0511	80.4%	1545.0
DeepSeek-VL2-Tiny [32]	Static VLM	62.57	285.5	7.3	0.144	$0.0143	94.5%	1991.0
Offline-Linear	Offline Router	78.21	384.3	106.2	1.721	$0.1777	31.8%	1036.0
KNN-Router (k=15)	Offline Router	78.31	361.8	83.7	1.444	$0.1306	49.9%	1030.0
ARMS-Router [30]	Offline Router	78.80	369.0	90.8	1.306	$0.2190	16.0%	1000.0
MLP-Router [14]	Offline Router	79.85	400.4	122.2	1.844	$0.1854	28.8%	936.0
FrugalGPT-Cascade [5]	Seq. Cascade	73.23	725.8	69.7	1.719	$0.1704	34.6%	1340.0
Epsilon-Greedy (ϵ=0.1)	Online Bandit	69.25 ± 3.74	364.2	86.0	1.445	$0.2803	-7.6%	1583.2
UCB1 [3]	Online Bandit	68.42 ± 0.75	321.6	43.4	1.271	$0.1622	37.7%	1634.0
Thompson Sampling [2]	Online Bandit	69.50 ± 0.86	340.3	62.2	1.396	$0.1595	38.8%	1568.0
DQN-Router [24]	Online Bandit	72.96 ± 1.02	335.5	57.3	1.421	$0.1845	29.2%	1356.4
MHR-Router [36]	Retrieval Bandit	71.48 ± 0.51	346.4	68.2	1.333	$0.1247	52.1%	1446.8
LinUCB (Cold Online)	Online Bandit	72.68 ± 1.71	363.0	84.9	1.535	$0.3220	-23.6%	1373.8
CascadeRoute-VL (Frugal γ=0.30)	Proposed	76.13 ± 0.21	360.5	51.1	0.824	$0.0876	66.4%	1162.8
CascadeRoute-VL (Full, γ=0.08)	Proposed	78.59 ± 0.12	370.8	91.9	1.550	$0.1598	38.7%	1013.0
Oracle (Hindsight 17-Arm)	Upper Bound	95.17	296.5	18.3	0.339	$0.0348	86.6%	0.0

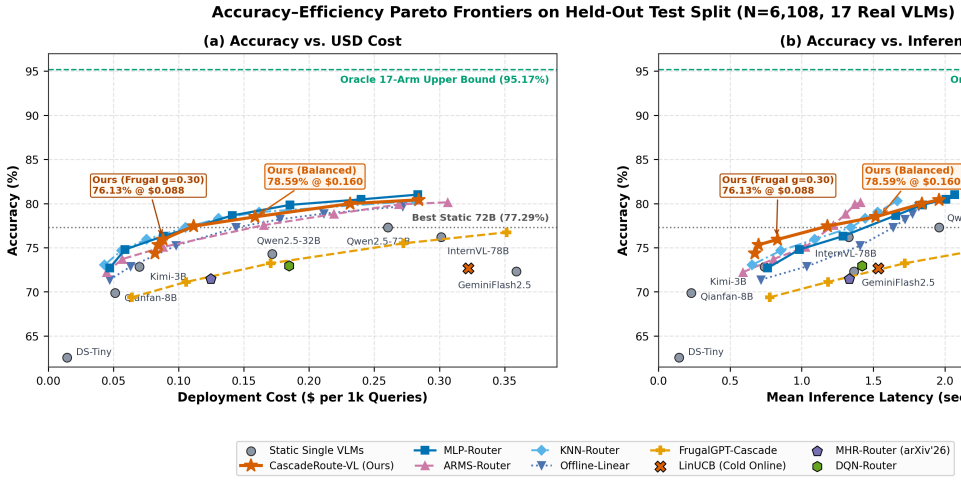


Figure 3. Accuracy vs. Cost and Inference Latency Pareto Frontiers on Held-Out Test (N=6,108). Sweeping $\gamma \in [0, 0.9]$ shows CascadeRoute-VL spans a monotone frontier exceeding monolithic 72B while staying cheaper and faster than offline routers and FrugalGPT.

Table 2. Zero-Leakage OOD Distribution Shift (General+STEM → Unseen Charts+OCR+Hall, N=17,601, 5 Seeds). Supervised offline routers collapse under shift, whereas CascadeRoute-VL adapts online to 80.68%.

Method	In-Dist	OOD	Shift Δ	Exceeds MLP
Qwen2.5-VL-72B (Static)	69.24	83.81	+14.56	Ref.
Offline-Linear	73.64	65.41	-8.24	Never
ARMS-Router [30]	76.06	66.58	-9.48	Never
MLP-Router [14]	72.77	56.97	-15.80	Baseline
LinUCB (Cold Online)	74.44	75.72	+1.28	∼191 Q
CascadeRoute-VL (Ours)	75.98	80.68	+4.70	∼51 Q
Oracle (Upper Bound)	95.48	95.36	-0.12	Instant

Table 3. Component & Zero-Label Modality Ablations (Test N=6,108, 5 Seeds). Warm-start cuts cold exploration cost; perceptual cues lift cold LinUCB by +3.86%.

Ablation Variant	Acc (%)	Std	Cost / 1k	Cold Acc
(a) Progressive Component Additions (γ=0.08)				
1. Pure Online LinUCB (Cold)	72.68	1.71	$0.3220	—
2. + Offline Warm-Start Prior	78.58	0.11	$0.1590	—
3. + Budget Annealing ($\alpha_t = \alpha_0 b_t$)	78.59	0.14	$0.1591	—
4. + Confidence Gate (Full)	78.59	0.12	$0.1598	—
(b) Zero-Label Context Modalities (d dims)				
Text Only (MiniLM, 384-d)	77.82	0.23	$0.1532	70.50
Perceptual Cues Only (8-d)	77.82	0.07	$0.1804	66.51
Vision Only (DINO+ResNet, 512-d)	78.01	0.09	$0.1563	67.87
Vision + Perceptual (520-d)	77.92	0.06	$0.1555	70.52
Text + Vision Only (896-d)	78.77	0.16	$0.1605	68.82
Full Joint Multimodal (904-d)	78.59	0.12	$0.1598	72.68

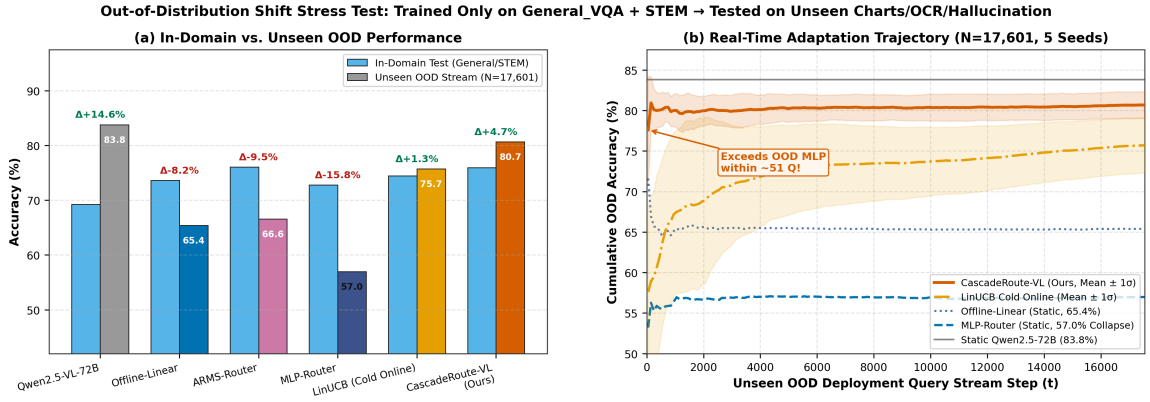


Figure 4. Out-of-Distribution Shift Stress Test (N_{OOD}=17,601, 5 Seeds). (a) Static offline routers collapse moving from General+STEM to unseen OCR/charts (MLP-Router: -15.80%, ARMS: -9.48%). (b) CascadeRoute-VL surpasses OOD MLP-Router within ∼51 queries and reaches 80.68% (+23.71% over static MLP).

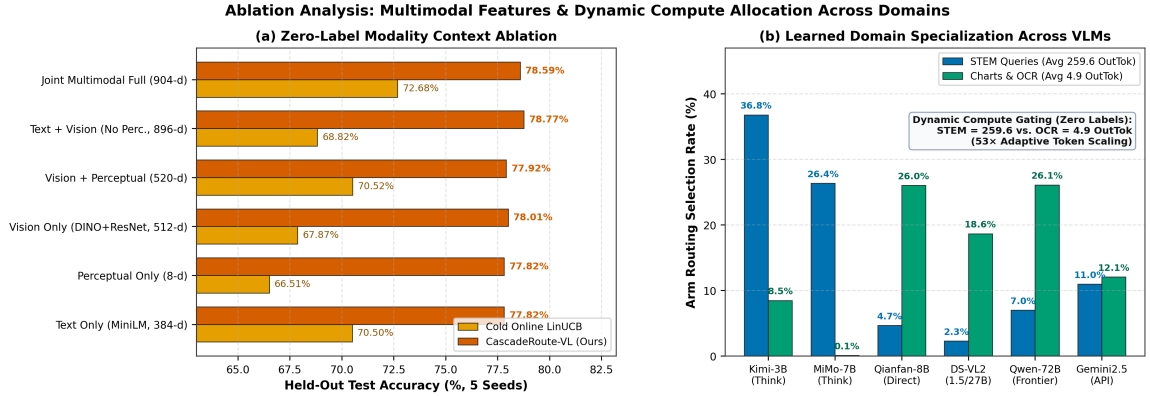


Figure 5. Modality Feature Ablations and Autonomous Domain Specialization (Test N=6,108). (a) Adding 8 unsupervised perceptual/linguistic cues boosts cold LinUCB from 68.82% to 72.68% (+3.86%) and lowers variance. (b) With zero domain labels, CascadeRoute-VL directs STEM traffic to thinking specialists and Charts/OCR to compact perception models (53× token shift: 259.6 vs. 4.9).

Table 4. Counterfactual Off-Policy Evaluation (Test N=6,108, 1,000 Bootstrap Resamples). 95% CIs for IPS, SNIPS, and Doubly Robust (DR) cleanly bracket true on-policy accuracy.

Policy (π)	True Acc	IPS (95% CI)	SNIPS (95% CI)	DR (95% CI)
DS-VL2-Tiny	0.6257	0.631 [0.520, 0.740]	0.592 [0.526, 0.663]	0.576 [0.516, 0.643]
Qianfan-VL-8B	0.6988	0.729 [0.615, 0.845]	0.678 [0.616, 0.742]	0.696 [0.631, 0.759]
Qwen2.5-VL-72B	0.7729	0.844 [0.722, 0.971]	0.856 [0.810, 0.903]	0.821 [0.778, 0.866]
Offline-Linear	0.7883	0.782 [0.759, 0.805]	0.784 [0.770, 0.798]	0.785 [0.771, 0.799]
MLP-Router	0.7985	0.760 [0.665, 0.862]	0.780 [0.729, 0.827]	0.774 [0.724, 0.820]
CascadeRoute-VL	0.7878	0.836 [0.733, 0.937]	0.792 [0.746, 0.839]	0.783 [0.740, 0.828]

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